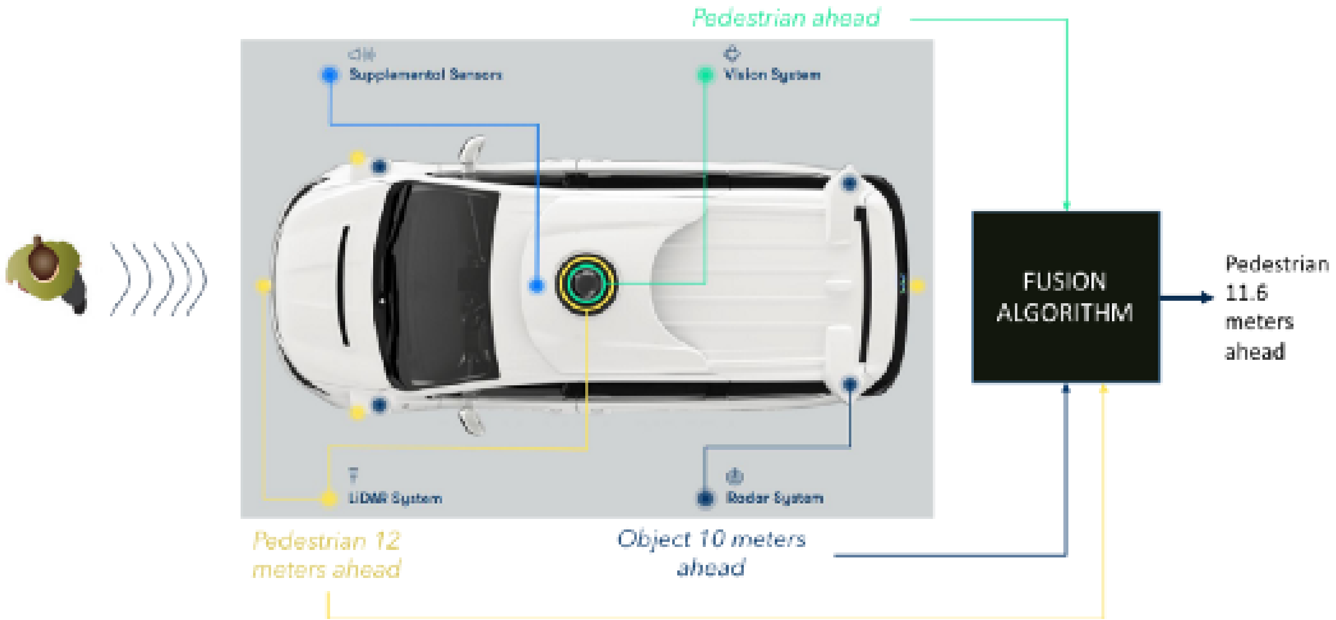
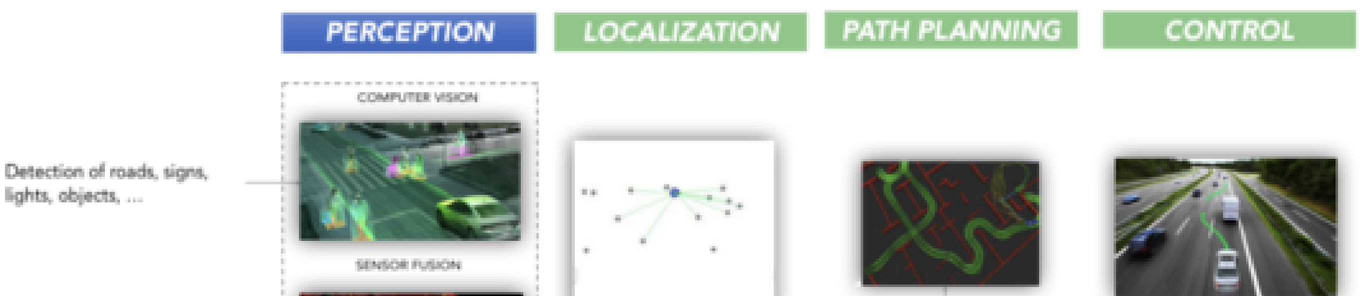
Sensor Fusion

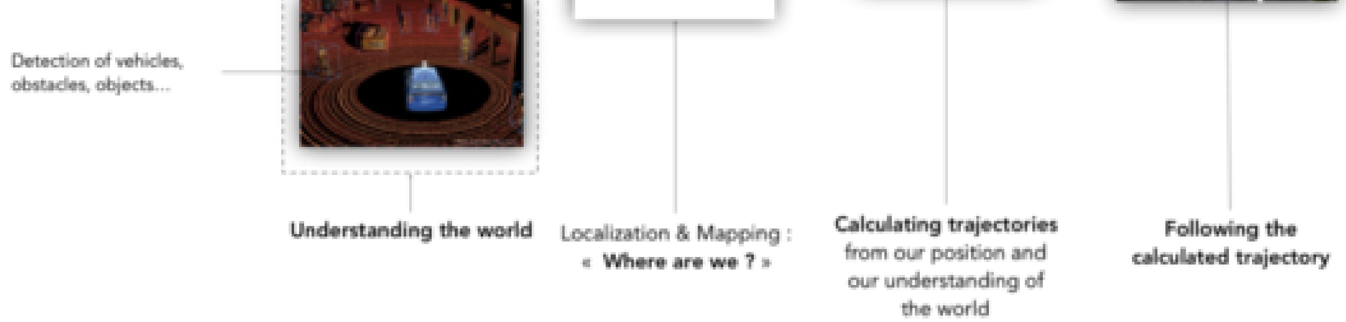
Jeremy Cohen May 22, 2018 · 10 min read



Sensor fusion is a crucial step for autonomous vehicles. How are the sensors used in autonomous vehicles?

> *This article follows the article IA … And the vehicle went autonomous! on computer vision. This article evoked a diagram describing the operation of an autonomous vehicle :*





How self-driving cars work

We mentioned here four major steps in the operation of an autonomous vehicle. After being interested in computer vision, let's move on to **Sensor Fusion**.

> *This discipline is used in autonomous vehicles in the **Perception** step, used to understand the world around our car by combining the sensors present.*

Before we start, I invite you to join the Think Autonomous Mailing List and learn every day about self-driving cars, Computer Vision, and Artificial Intelligence.

Sensors

The autonomous vehicle uses a large number of sensors to understand its environment, to locate and move.

Different sensors on a self-driving car

Camera

The camera transcribes the driver's vision. It is very often used to understand the environment with artificial intelligence by classifying roads, pedestrians, signs… as discussed in the previous article …

Vehicle detection using a camera

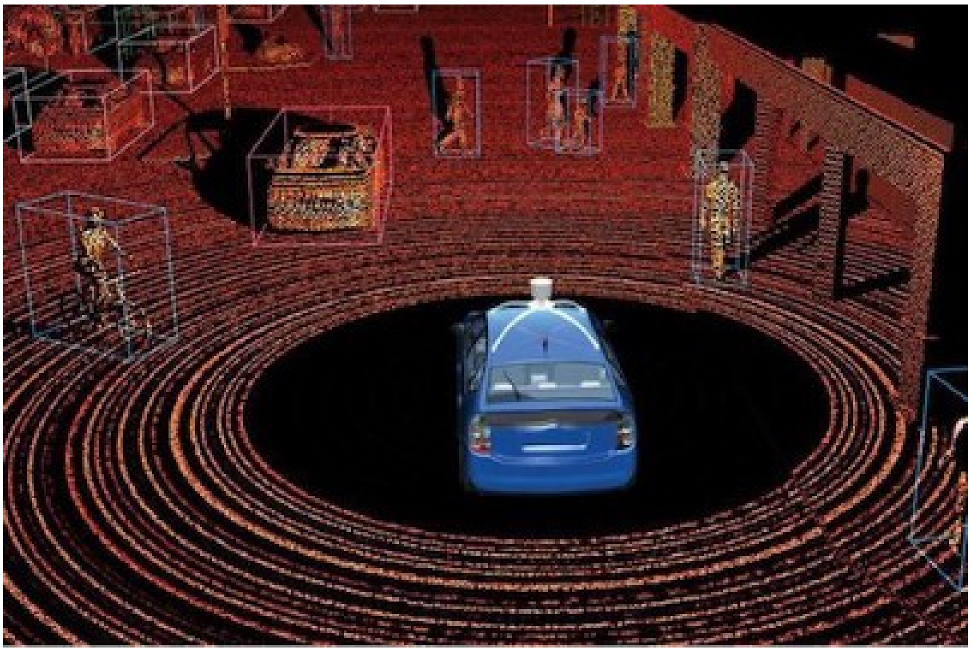
Radar (Radio Detection and Ranging)

The radar emits radio waves to detect objects within a radius of several meters. Radars have been in our cars for years to detect vehicles in blind spots or to avoid collisions. They have better results on moving objects than on static objects.
Unlike other sensors that calculate the difference in position between two measurements, the radar uses the Doppler effect by measuring the change in the next wave frequency if the vehicle moves towards us or moves away. This is called radial velocity.

> ***The radar can directly estimate a speed.***
> *It has a low resolution, it allows to know the position and speed of a detected object. On the other hand, it can not determine what is the object being sensed.*

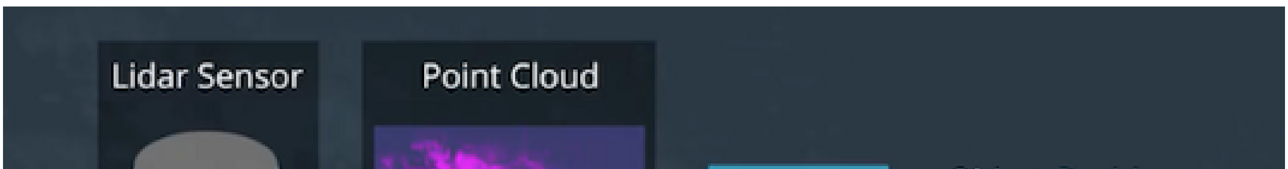
Lidar (Light Detection and Ranging)

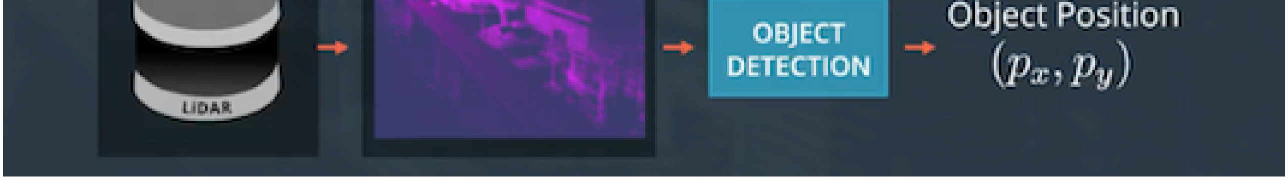
The lidar uses infrared sensors to determine the distance to an object. A rotating system makes it possible to send waves and to measure the time taken for this wave to come back to it. This makes it possible to generate a points cloud of the environment around the sensor. A lidar can generate about 2 million points per second. This point cloud giving different 3D shapes, **it is possible to make the classification of objects thanks to a Lidar**.



Point cloud of the environment

It allows a great distance (100 to 300 m) to estimate the position of objects around him. Its size is however cumbersome since it exceeds the roof of the vehicles. Not mentioned, its price can go up to 100,000.





Position estimation with Lidar

Ultrasonic sensors

Ultrasonic sensors are used to estimate the position of static vehicles, for example, to park. They are much cheaper but have a range of a few meters.

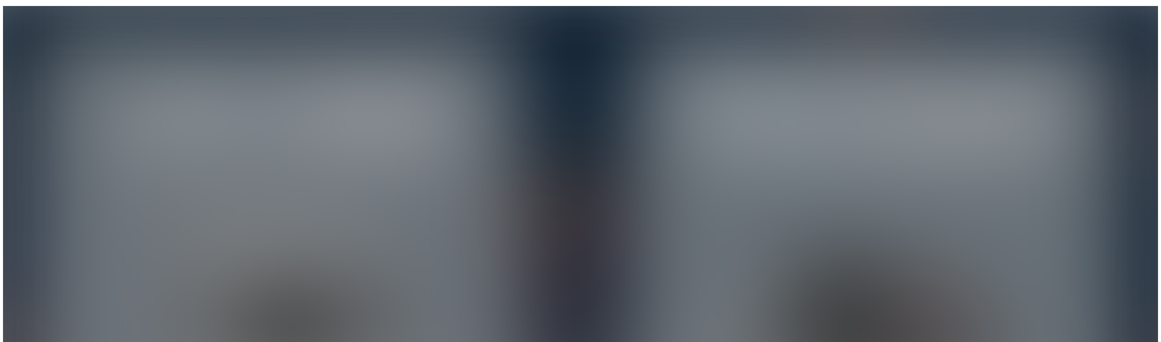
Odometric sensors

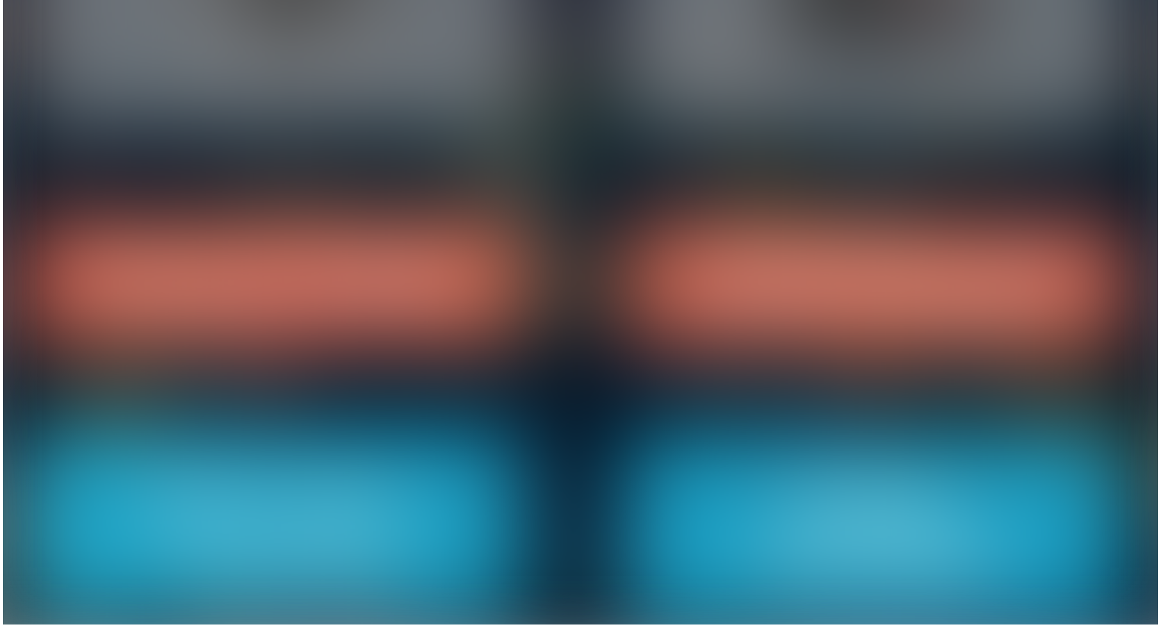
They make it possible to estimate the speed of our vehicle by studying the displacement of its wheels.

Benchmark

Sensor Benchmark

> *Each of these sensors has advantages and disadvantages. The aim of **sensor fusion** is to use the advantages of each to precisely understand its environment. The camera is a very good tool for detecting roads, reading signs or recognizing a vehicle. The Lidar is better at accurately estimating the position of this vehicle while the Radar is better at accurately estimating the speed.*





Better use of Radar and Lidar

In the cycle of operation of an autonomous vehicle, there is Perception. This step is crucial for the vehicle to detect obstacles, roads, pedestrians … A vehicle is full of sensors, so it is very important to understand how these sensors are used to understand how a car can drive without a driver.

Kalman filters

> *The algorithm used to merge the data is called a* ***Kalman filter****.*

The Kalman filter is one of the most popular algorithms in data fusion. Invented in 1960 by Rudolph Kalman, it is now used in our phones or satellites for navigation and tracking. The most famous use of the filter was during the Apollo 11 mission to send and bring the crew back to the moon.

When to use a Kalman Filter ?

> *A Kalman filter can be used for data fusion to estimate the state of a dynamic system (evolving with time) in the present (filtering), the past (smoothing) or the future (prediction). Sensors embedded in autonomous vehicles emit measures that are sometimes*

> *incomplete and noisy. The inaccuracy of the sensors (noise) is a very important problem and can be handled by the Kalman filters.*

A Kalman filter is used to **estimate the state of a system**, denoted **x**. This vector is composed of a position p and a velocity v.

State of a system

At each estimate, we associate a measure of **uncertainty P**.

By performing a fusion of sensors, we take into account different data for the same object. A radar can estimate that a pedestrian is 10 meters away while the Lidar estimates it to be 12 meters. The use of Kalman filters allows you to have a precise idea to decide how many meters really is the pedestrian by eliminating the noise of the two sensors.

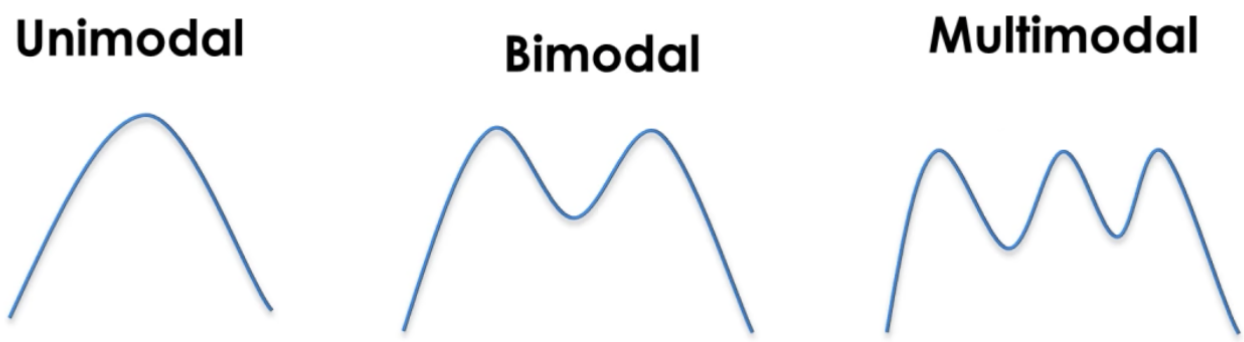
> A Kalman filter can generate estimates of the state of objects around it. To make an estimate, it only needs the current observations and the previous prediction. The measurement history is not necessary. This tool is therefore light and improves with time.

How it looks like

State and uncertainty are represented by Gaussians.

Gaussian

A Gaussian is a continuous function under which the area is 1. This allows us to represent probabilities. We are on a **probability to normal distribution**. The uni-modality of the Kalman filters means that we have a single peak each time to estimate the state of the system.



unimodal vs multimodal (source)

We have a **mean μ** representing a state and a **variance σ^2** representing an uncertainty. The larger the variance, the greater the uncertainty.

Different gaussians

> Gaussians make it possible to estimate probabilities around the state and the uncertainty of a system. A Kalman filter is a continuous and uni-modal function.

Bayesian Filtering

> *In general, a Kalman filter is an implementation of a Bayesian filter, ie a sequence of alternations between prediction and update or correction.*

Bayes filter

> ***Prediction**: We use the estimated state to predict the current state and uncertainty.*
> ***Update**: We use the observations of our sensors to correct the predicted state and obtain a more accurate estimate.*

> To make an estimate, a Kalman filter only needs current observations and the previous prediction. The measurement history is not necessary.

Maths

The mathematics behind the Kalman filters are made of additions and multiplications of matrices. We have two stages: **Prediction** and **Update**.

Prediction

Our prediction consists of estimating **a state x'** and **an uncertainty P'** at time **t** from the previous states x and P at **time t-1**.

Prediction formulas

- F: Transition matrix from t-1 to t
- ν: Noise added
- Q: Covariance matrix including noise

Update
The update phase consists of using a **z measurement** from a sensor to correct our prediction and thus **predict x and P**.

Update formulas

- y: Difference between actual measurement and prediction, ie the error.
- S: Estimated system error

- H: Matrix of transition between the marker of the sensor and ours.

- R: Covariance matrix related to sensor noise (given by the sensor manufacturer).

- K: Kalman gain. Coefficient between 0 and 1 reflecting the need to correct our prediction.

The update phase makes it possible to estimate an x and a P closer to reality than what the measurements provide.

> A Kalman filter allows predictions in real time, without data beforehand. We use a mathematical model based on the multiplication of matrices for each time defining a state x (position, speed) and uncertainty P.

Représentation Prior/Posterior

This diagram shows what happens in a Kalman filter.

Kalman filter estimation (source)

- **Predicted state estimate** represents our first estimate, our prediction phase. We are talking about prior.

- **Measurement** is the measurement from one of our sensors. We have better uncertainty but the noise of the sensors makes it a measurement that is always difficult to estimate. We talk about likelihood.

- **Optimal State Estimate** is our update phase. The uncertainty is this time the weakest, we accumulated information and allowed to generate a value more sure than with our sensor alone. This value is our best guess. We speak of posterior.

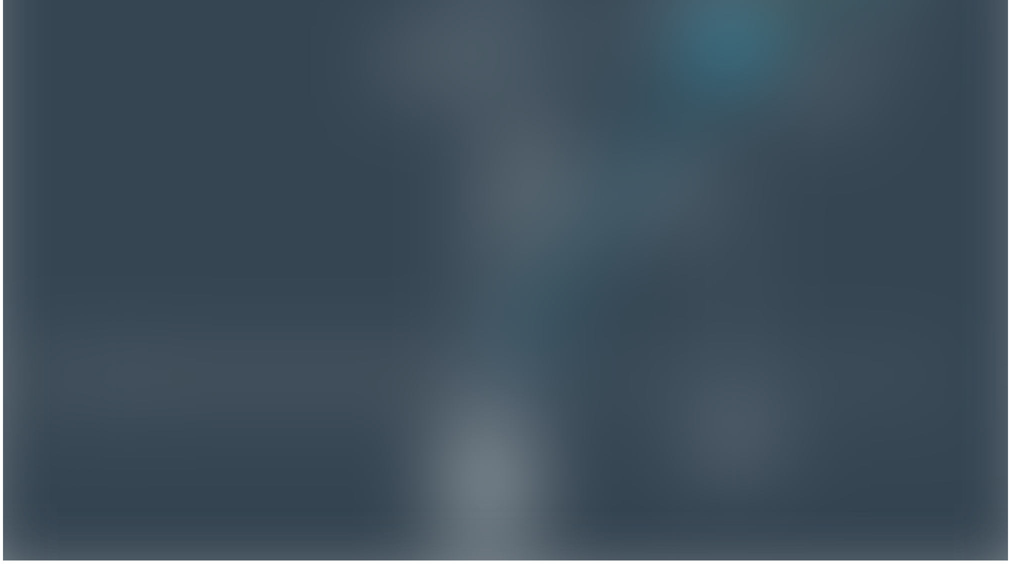
What a Kalman filter implements is actually a Bayes rule.

Bayes rule

> In a Kalman filter, we loop predictions from measurements. Our predictions are always more precise since we keep a measure of uncertainty and regularly calculate the error between our prediction and reality. We are able from matrix multiplications and probability formulas to estimate velocities and positions of vehicles around us.

“Extended/Unscented” filters and non-linearity

An essential problem arises. Our mathematical formulas are all implemented with linear functions of type y = ax + b.
A Kalman filter always works with linear functions. On the other hand, when we use a Radar, the data is not linear.



Functionment of a Radar

The Radar sees the world with three measures:

- **ρ (rho)**: The distance to the object that is tracked.
- **φ (phi)**: The angle between the x-axis and our object.
- **$\dot{\rho}$ (rhodot)**: The change of ρ, resulting in a radial velocity.

These three values make our measurement a nonlinear given the inclusion of the angle **φ**.

Our goal here is to convert the data **ρ, φ, $\dot{\rho}$** to Cartesian data (px, py, vx, vy).

If we enter non-linear data in a Kalman filter, our result is no longer in uni-modal Gaussian form and we can no longer estimate position and velocity.

Gaussian vs non-linearity

> *So we use approximations, which is why we are working on two methods:*
> *- **Extended Kalman filters** use Jacobian and Taylor series to linearize the model.*
> *- **Unscented Kalman filters** use a more precise approximation to linearize the model.*

> To deal with the inclusion of non linearity by the Radar, techniques exist and allow our filters to estimate position and velocity of the objects that we wish to track.

Results

The following videos present the results for an extended Kalman filter and an unscented kalman filter. We can see in particular our **sensor (black origin)**, the **Radar data (red)**, the **Lidar data (blue)** and our **prediction (green)**. Each time we receive some data from one of our sensor, we realize a prediction.

Extended Kalman Filter

Unscented Kalman Filter

RMSE values are *Root Mean Squared Error* measures, which is the error between our prediction and the reality.
They are lower for the unscented filter than extended because this technique is more efficient.

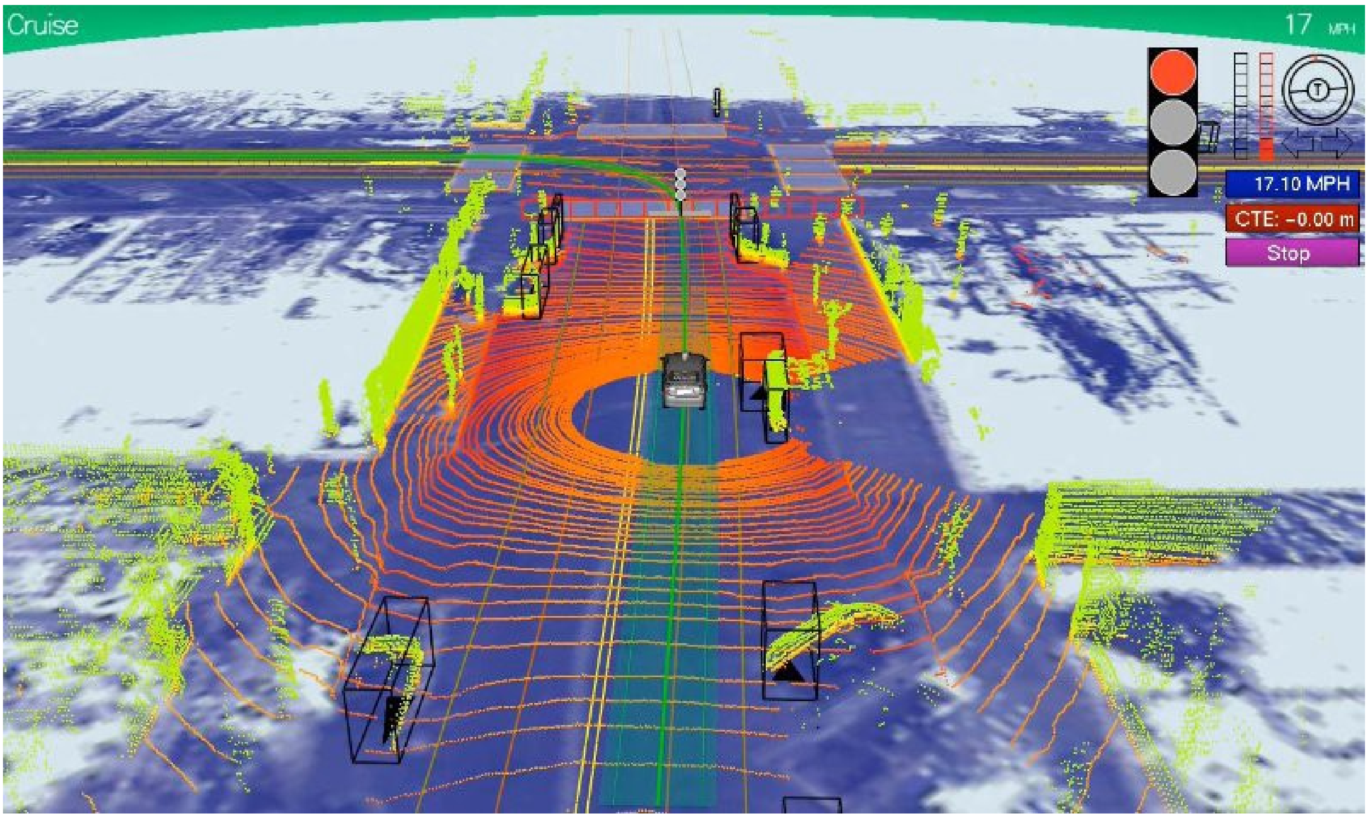
Conclusion

> Sensor fusion is one of the most important topics in the field of autonomous vehicles. Fusion algorithms allow a vehicle to understand exactly how many obstacles there are, to estimate where they are and how fast they are moving.
>
> Depending on the sensor used, we can have different implementations of the Kalman Filter.

Sensor Fusion general flux for Radar and Lidar

Kalman filters are used in smartphones, satellites, and navigation systems to estimate the state of a system. Very popular and used, data fusion algorithms now make vehicles autonomous.

Before I conclude, I would like to invite you to the private mailing list. Every day, I send an email where I share my experience on AI and Autonomous technology. Starting tomorrow, you will receive life-changing career advices, technical content like this one or discounts for my courses. Subscribe here, I hope to see you in tomorrow's email.



Jeremy Cohen.

🚀 Receive daily emails on self-driving cars and AI and lead the Autonomous Tech world!

> *Discover this article in French*

Next on Localization, Path Planning, Control and Final Integration

References

- Udacity Self-driving Car Nanodegree Program

Sensor Fusion GitHub Projects

- Implementation of an extended Kalman Filter : extended-kalman-filters
- Implementation of an unscented Kalman Filter : unscented-kalman-filters

Self Driving Cars Lidar Radar Kalman Filter Artificial Intelligence